Online Food Delivery Analysis: Data-Driven Business Insights

Customer & Order Analysis

11.Cancellation rate by restaurant.

No restaurant has a genuinely high cancellation rate. The ten shown cancel about 21 to 22% of orders against the overall 15.0%, but that gap is what random variation produces across 500 restaurants. The correlation between a restaurant's cancellation rate and its average rating is -0.06.

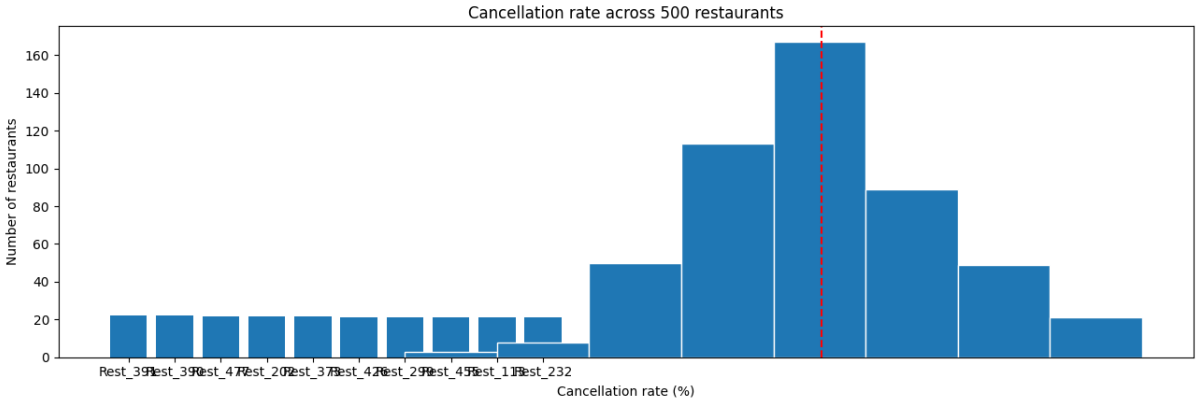
Illustrative sample records:

Restaurant Name	orders	cancelled	cancel rate
Restaurant_391	196	43	21.9
Restaurant_390	201	44	21.9
Restaurant_477	176	38	21.6
Restaurant_202	204	44	21.6
Restaurant_373	191	41	21.5
Restaurant_426	212	45	21.2
Restaurant_299	194	41	21.1
Restaurant_455	180	38	21.1
Restaurant_113	205	43	21.0
Restaurant_232	177	37	20.9

2 . VISUALIZATIONS

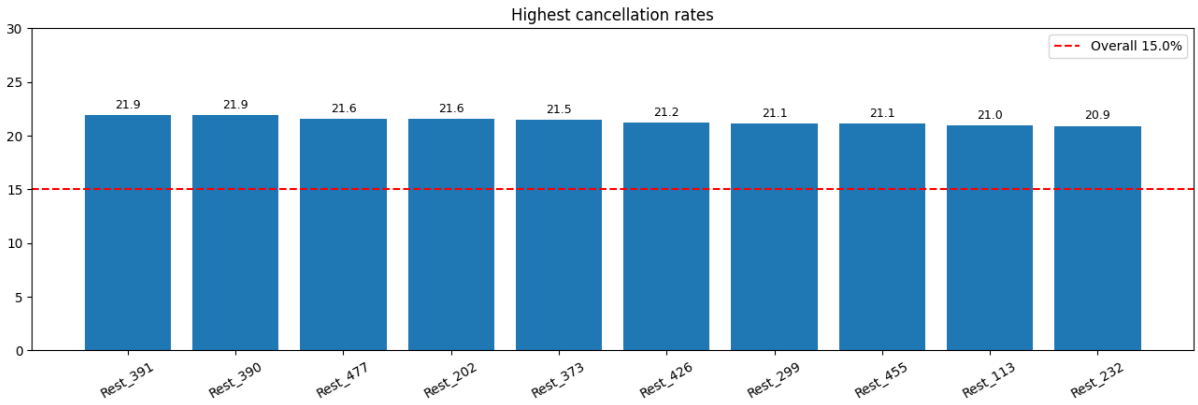
Visualization – 1

This histogram shows how the 500 restaurants' cancellation rates are spread.



Visualization – 2

This bar chart shows high cancellation rates in top 10 restaurant.



3.ANALYTICAL INSIGHTS – DEMAND AND REVENUE

INSIGHT – 1: Cancellation rates vary but cluster around 15%.

The overall rate is 15.0%. The middle 80% of restaurants fall between

11.7% and 18.5%, and the extremes run from 7% to 22%. A typical restaurant is not far from the average.

INSIGHT – 2: Each percentage point of cancellations costs about ₹1.7M.

Spread across the platform, cutting the cancellation rate from 15% to 14% is worth about ₹1.7M in recovered sales.

INSIGHT – 3: Cancellations cost 15% of potential sales.

On top of the ₹143.1M earned from delivered orders, another ₹25.2M was lost to cancellations. That's about ₹50K per restaurant on average, ranging from ₹18K to ₹85K.

INSIGHT – 4: The worst restaurants are not the main source of losses.

The 50 restaurants with the highest rates account for only 13.1% of cancellations and 12.9% of lost revenue. Most of the lost sales come from the average restaurants.

4.ANALYTICAL INSIGHTS-OPERATIONS AND CUSTOMER EXPERIENCE:

INSIGHT-5: Identify high cancellation rate by restaurant

Identify restaurants with unusually high cancellation rates and prioritize corrective actions.

INSIGHT-6: Improve restaurant rejection rate

Analyse how often restaurants reject incoming orders and identify common rejection reasons. To decrease rejection we have to ensure restaurants accept orders only when they can fulfil them.

INSIGHT-7: Estimate cancelled order value

Prioritize improvements based on financial impact, not cancellation rate alone.